

MATH8009: Matrix Operations - Problem Sheet 8

Q1.

(i) (a) $\begin{pmatrix} 7 & 3 \\ 1 & 9 \\ -2 & 15 \end{pmatrix}$ (b) $\begin{pmatrix} 9 & -14 \\ 7 & -7 \\ -9 & -20 \end{pmatrix}$ (c) *Not defined*

(ii) (a) $\begin{pmatrix} 9 & -1 \\ -1 & 1 \end{pmatrix}$ (b) $\begin{pmatrix} -2 & 8 \\ -12 & 2 \end{pmatrix}$ (c) $\begin{pmatrix} 22 & -10 \\ -10 & 6 \end{pmatrix}$

(iii) (a) $\begin{pmatrix} 7 \\ 2 \\ 2 \end{pmatrix}$ (b) $\begin{pmatrix} 4 \\ -21 \\ 24 \end{pmatrix}$ (c) *Not defined*

Q2. $C = \begin{pmatrix} 6 & 16 & -22 \\ -28 & -14 & -2 \end{pmatrix}$

Q3. $C = \begin{pmatrix} 22 & 55 \\ -8 & 27 \end{pmatrix}$ and $D = \begin{pmatrix} -40 & 66 \\ -12 & 18 \end{pmatrix}$

Q4 (i) *Not defined*, (ii) *Defined*, 4×2 , (iii) *Not defined*, (iv) *Not defined*,
(v) *Defined*, 5×2 , (vi) *Defined* 5×4

Q5. Solve for x, y and z in the matrix equation

$$4 \begin{pmatrix} x & y \\ z & -1 \end{pmatrix} = 2 \begin{pmatrix} y & z \\ -x & 1 \end{pmatrix} + 2 \begin{pmatrix} 4 & x \\ 5 & -x \end{pmatrix}.$$

$$4x = 2y + 8$$

$$4y = 2z + 2x$$

$$4z = -2x + 10$$

$$-4 = 2 - 2x$$

$$x = 3, y = 2, z = 1$$

Q6. Solve for x, y, z and w in the matrix equation

$$\begin{pmatrix} w & x \\ y & x \end{pmatrix} = \begin{pmatrix} -4 & 3 \\ 2 & -1 \end{pmatrix} + 2 \begin{pmatrix} y & w \\ z & x \end{pmatrix}.$$

$$w = -4 + 2y$$

$$x = 3 + 2w$$

$$y = 2 + 2z$$

$$x = -1 + 2x$$

$$x = 1, y = \frac{3}{2}, z = -\frac{1}{4}, w = -1$$

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Q8. (a) $\begin{pmatrix} 0.2 & -0.6 \\ 0 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 0.25 & 0.5 \\ -0.125 & 0.25 \end{pmatrix}$

(c) *Does not exist*

Q9. (a) $\begin{pmatrix} -0.5 & 2 \\ 0.5 & -1 \end{pmatrix}$

(b) $\begin{pmatrix} 2 & 3 & -15 \\ -4 & -7 & 34 \\ 1 & 2 & -9 \end{pmatrix}$

(c) $\begin{pmatrix} -7/3 & 1/3 & 1/3 \\ -5/3 & -1/3 & 2/3 \\ 1 & 0 & 0 \end{pmatrix}$